PRISM-XAI: Privacy-Accounted Semantic Explanation Scheduling for Smart-Grid Edge Intelligence

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***Abstract*—Repeatedly delivering household-level explanations can help a grid operator understand rapid demand changes, but explanation packets consume wireless resources and may reveal private behaviour. We introduce *PRISM-XAI*, a two-layer design for privacy-accounted semantic explanation delivery. The first layer, Multi-level Adaptive Privacy for XAI (MAP-XAI), converts explanation changes into compact private semantic packets: an endpoint clips an attribution delta, applies a Gaussian differential-privacy (DP) mechanism, and only then performs Top-k selection and quantization while a Rényi-DP accountant tracks cumulative household privacy. The second layer, PRISM-XAI, decides whether, when, from which endpoint, and at what resolution those packets should be transmitted using channel state, packet deadline, Age of Explanation (AoX), public grid criticality, endpoint fairness, residual signal, and expected fidelity gain per bit. In 720 paired runs on two public electricity-demand datasets, two attribution methods, three privacy budgets, six schedulers, and 10 seeds, the proposed stack incurs no privacy-budget violations. At the selective-release operating point, PRISM-XAI reduces explanation traffic by 79.4% relative to joint full-packet delivery and by about 94.6% relative to fixed-DP schedulers, with significantly lower traffic in all 48 paired external-baseline comparisons. A feasibility ladder shows that Top-K overlap remains 0.990/0.993 after clipping but falls to 0.241/0.282 after cumulative DP; subsequent compression changes it only to 0.252/0.295. Thus, the dominant fidelity bottleneck is repeated household-level privacy expenditure, not semantic compression, motivating selective private release rather than further compression-only tuning.**

***Index Terms*—smart-grid communications, edge intelligence, explainable AI, differential privacy, semantic communications, resource allocation, Age of Explanation**

I. Introduction

Short-term load forecasts are more useful when an operator can also see which factors drove the prediction. A demand surge caused by heating may require a different response from one driven by electric-vehicle charging or an unusual subset of customers. Feature-attribution methods such as SHAP [3] and Integrated Gradients [4] can provide this evidence, and explainable AI (XAI) is increasingly studied for load forecasting and grid monitoring [12]–[14], [16].

Repeated explanation delivery creates a communication–privacy problem that is different from evaluating XAI offline. First, household attributions can leak information about private behaviour; explanations are themselves potential privacy channels [17]. Second, transmitting every explanation in full consumes scarce uplink capacity and may miss an operational deadline. Third, privacy loss composes over repeated releases. A mechanism that is private per packet can still exhaust a household's privacy allowance over an operating horizon. Finally, the value of a packet is time varying: a stale explanation during a critical grid condition can be more important than a fresh explanation during routine operation.

We therefore separate two questions that are often conflated: *how should a private explanation packet be represented?* and *when should that packet be transmitted?* Our first contribution, MAP-XAI, answers the first question. It provides a compact, hierarchy-aware semantic packet under cumulative privacy accounting. Our second contribution, PRISM-XAI, answers the second question. It treats a MAP-XAI packet as a valid semantic mode within a wider scheduler that may instead use a richer packet, a coarse zonal output, or no private transmission. This relationship is important: MAP-XAI is not an external competitor to PRISM-XAI, but a proposed packetization layer and an internal restricted-mode ablation of the full system.

The contributions are:

- We introduce MAP-XAI, a private semantic XAI packetization layer that clips explanation deltas, applies Gaussian DP *before* Top-k/quantization, and composes household privacy with Rényi-DP.
- We introduce PRISM-XAI, an online wireless scheduler over MAP-XAI and alternative semantic modes. It combines AoX, public criticality, endpoint fairness, residual accumulation, explicit abstention, and expected explanation gain per transmitted bit under radio, deadline, and remaining-privacy constraints.
- We evaluate the complete stack on two real load datasets using 720 paired runs and four external scheduling baselines. We report communication, explanation, operational, fairness, latency, and privacy metrics rather than a single fidelity score.
- We add a feasibility ladder and privacy–release sweep that isolate where explanation signal is lost. The diagnostics show that cumulative household-level DP, rather than clipping or semantic compression, is the main limitation and reveal the resulting coverage–fidelity trade-off.

II. Related Work

Power-system XAI has been used for load forecasting, integrated-energy forecasting, frequency analysis, and anomaly detection [12]–[15]. These works mainly study explanation quality after prediction; they do not jointly schedule repeated explanation packets over a time-varying wireless edge. Privacy-preserving and federated XAI has highlighted

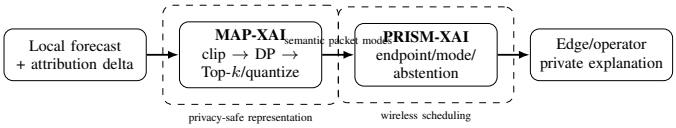


Fig. 1. Proposed stack. MAP-XAI determines how a private semantic explanation is represented; PRISM-XAI determines whether, when, and at what resolution it is delivered.

explanation leakage, communication cost, continual release, and non-identically distributed clients as open issues [6]. Related adaptive-DP approaches manage noise or accumulated privacy during distributed learning [7], [8]. Our setting differs because the forecasting model can already be deployed; the scarce resources are operator-facing explanation packets, wireless capacity, deadline slack, and each household's cumulative privacy allowance.

Age of Information (AoI) captures freshness of status updates [5], [18]. We use *Age of Explanation* (AoX): the age of the last accepted explanation, weighted by public grid criticality. Semantic communications emphasizes task-relevant meaning rather than faithful reproduction of every source symbol [19]. Here the semantics are explicit and auditable: selected attribution indices, privatized changes, signs, resolution, and reliability metadata. Endpoint fairness is assessed with Jain's index [20].

III. Two-Layer Private Explanation Delivery

A. System and threat model

Let $\mathcal{H} = \{1, \dots, N\}$ denote household endpoints grouped into zones $\mathcal{H}_z$. At slot t, endpoint i observes $\boldsymbol{x}_{i,t} \in \mathbb{R}^d$, a deployed forecaster f produces $\hat{y}_{i,t}$, and an explainer produces $\boldsymbol{\phi}_{i,t} = \mathcal{E}(f, \boldsymbol{x}_{i,t}) \in \mathbb{R}^d$. Raw features and non-private attributions remain local. The trusted endpoint privatizes a packet before exposing it to the edge; the edge and operator may observe all transmitted packet contents and scheduling outcomes. Timing side-channels are outside the formal content-DP guarantee and require fixed opportunities or dummy traffic in timing-sensitive deployments.

B. MAP-XAI: private semantic packetization

Let t_i^- be the last accepted explanation slot. The endpoint forms $\Delta\boldsymbol{\phi}_{i,t} = \boldsymbol{\phi}_{i,t} - \boldsymbol{\phi}_{i,t_i^-}$ and clips it to ℓ_2 radius C_i:

$$\overline{\Delta\boldsymbol{\phi}}_{i,t} = \frac{\Delta\boldsymbol{\phi}_{i,t}}{\max\{1, \|\Delta\boldsymbol{\phi}_{i,t}\|_2 / C_i\}}. \tag{1}$$

For add/remove household adjacency, sensitivity is $\Delta_i = C_i$. A Gaussian mechanism then releases

$$\widetilde{\Delta\boldsymbol{\phi}}_{i,t} = \overline{\Delta\boldsymbol{\phi}}_{i,t} + \boldsymbol{n}_{i,t}, \quad \boldsymbol{n}_{i,t} \sim \mathcal{N}(0, \sigma_{i,t}^2 \Delta_i^2 \boldsymbol{I}_d). \tag{2}$$

Only after Eq. (2) does MAP-XAI retain Top-k coordinates and quantize their magnitudes. Hence selection and quantization are DP post-processing. A mode $m = (k, q, \sigma)$ has packet size

$$S_i(m) = S_0 + k(\lceil \log_2 d \rceil + q + 1), \tag{3}$$

where the final bit carries the sign. Different k, q, and σ values define semantic/privacy levels; a no-release action spends no privacy.

We account for repeated release with Rényi-DP (RDP) [2]. At order α, a Gaussian release costs $\rho_{i,t}(\alpha) = \alpha/(2\sigma_{i,t}^2)$. The endpoint composes these costs and rejects a private release that would exceed the remaining budget. At the end of the protected horizon,

$$\epsilon_i(\delta_i) = \min_{\alpha} \left\{ \sum_t a_{i,t} \rho_{i,t}(\alpha) + \frac{\log(1/\delta_i)}{\alpha - 1} \right\}. \tag{4}$$

Privacy is charged when a private packet is exposed outside the trusted endpoint, even if it later misses the communication deadline. Because clipping fixes the sensitivity, Eq. (2) is Gaussian DP at each release; Top-k selection, quantization, and packet reconstruction are post-processing. RDP composition in Eq. (4) therefore provides the stated user-level horizon guarantee whenever every admitted release remains inside the tracked RDP budget.

a) Content-privacy guarantee.: Under the add/remove household adjacency above, each exposed noisy delta has order-α RDP cost $\rho_{i,t}(\alpha) = \alpha/(2\sigma_{i,t}^2)$. Adaptive RDP composition is additive, so every packet-content transcript admitted by Eq. (4) is $(\epsilon_i(\delta_i), \delta_i)$-DP. Top-$k$ selection, quantization, reconstruction, and controller operations that depend only on public state and previously released private outputs are post-processing. This guarantee is for released content; an independent timing side channel remains outside the threat model stated above.

C. PRISM-XAI: privacy-accounted scheduling

Let $A_{i,t}$ be AoX and $w_{i,t}^{\mathrm{crit}} \geq 1$ a public criticality weight. A candidate mode must fit the available resource blocks, meet the $D_i^{\max}$ deadline, and pass RDP admission. The controller estimates $G_{i,t}(m)$, the expected reduction in explanation loss, from AoX, public criticality, channel state, mode resolution, remaining budget, and previously released privatized residual evidence; it never uses the non-private oracle online. Unsent residual evidence is accumulated and decayed, so a private release is considered only when the predicted semantic signal is large enough relative to the DP-noise floor.

For a feasible mode, define gain per transmitted bit

$$\Gamma_{i,t}(m) = \frac{G_{i,t}(m)}{S_i(m)}. \tag{5}$$

The nonzero candidate set includes a MAP-XAI packet mode, optional higher-resolution modes, and a coarse zonal mode; zero transmission is always available. A richer mode replaces the MAP-XAI mode only when its gain per bit exceeds the MAP-XAI value by a configured margin. The slot utility also rewards criticality-weighted AoX and under-served endpoints while penalizing energy and RDP shadow price. The resulting multiple-choice knapsack selects at most one mode per endpoint under the slot resource-block limit. A minimum critical-event coverage target and minimum Jain fairness are

included as service guardrails. This structure makes MAP-XAI the privacy-valid semantic layer and PRISM-XAI the resource-allocation layer above it. With M candidate modes per endpoint and B available resource blocks, the slot-level multiple-choice knapsack can be solved by dynamic programming in $O(NMB)$ time; in our experiments M and B are small constants. The online decision therefore does not require re-running the forecaster or a non-private explanation oracle at the edge.

IV. Evaluation

A. Data, wireless setting, and baselines

We use Open Power System Data (OPSD) household data and UCI ElectricityLoadDiagrams20112014 [9], [10]. Nine OPSD endpoints are grouped into three synthetic zones and 32 UCI endpoints into four synthetic zones. Each forecaster uses 12 lag, rolling-statistic, and calendar features. The tree model obtains RMSE/MAE $0.926/0.545$ on OPSD and $1.289/0.221$ on UCI. We evaluate two attribution implementations, TreeSHAP and feature ablation, to ensure the scheduler conclusions are not tied to one tree explainer.

The uplink abstraction uses eight 180 kHz resource blocks, 1 ms slots, 5 ms deadlines, 3.5 GHz carrier frequency, path loss, correlated shadowing, small-scale fading, edge delay, and bursty failures inspired by 3GPP TR 38.901 [11]. It is a reproducible scheduler-level stress trace, not a link-level field validation.

The main selective-release matrix uses 512 evaluation slots after 96 calibration slots, privacy budgets $\epsilon_{\max} \in \{8, 16, 32\}$ with $\delta = 10^{-5}$, at most eight private releases per endpoint, a 5% minimum critical-event coverage target, and 0.88 minimum Jain fairness. There are 2 datasets $\times$ 2 XAI methods $\times$ 3 privacy budgets $\times$ 6 schedulers $\times$ 10 paired seeds $= 720$ runs.

External baselines are periodic fixed-DP, channel-aware fixed-DP, AoX-aware fixed-DP, and joint scheduling with full, uncompressed private packets. We additionally report *MAP-XAI-only*, which is not an external baseline: it restricts our stack to the MAP-XAI packet mode and therefore isolates the incremental effect of full PRISM scheduling. Metrics include transmitted traffic, Top-K and cosine agreement, critical-event coverage and recall, intervention precision, false intervention, bits per correct intervention, deadline misses, Jain fairness, and spent privacy.

B. Feasibility diagnostics

A separate three-seed feasibility ladder at $\epsilon_{\max} = 4$ removes one transformation at a time: (A) oracle explanation, (B) clipping only, (C) clipping plus cumulative Gaussian DP with a full packet, and (D) DP plus semantic compression. A release sweep then varies $\epsilon \in \{2, 4, 8, 16, 32\}$ and the maximum private releases per endpoint $M \in \{4, 8, 16, 32, 64\}$. These diagnostics are not used to tune the reported 720-run matrix; they identify the mechanism that limits explanation utility.

V. Results

A. Communication efficiency and operational trade-off

Table I macro-averages the 12 dataset/XAI/privacy settings. The proposed stack operates at a deliberately selective point: mean critical-event coverage is 7.8%, versus 24.8% for joint full-packet scheduling and 87.5% for AoX fixed-DP. The traffic reduction must therefore be read jointly with coverage, precision, and bits per correct intervention rather than as a free throughput gain.

Figure 2 is the primary v19 result. Averaged condition-by-condition, PRISM transmits 94.6% less explanation traffic than the three fixed-DP schedulers and 79.4% less than joint full-packet scheduling. Paired 95% confidence intervals show lower traffic in all 48 external-baseline comparisons and lower bits per correct intervention in 45/48. Top-K is statistically tied in 37/48 comparisons, higher in eight, and lower in three; intervention precision is higher in 38/48. The result is therefore an efficiency claim rather than universal recall dominance: AoX fixed-DP serves many more critical events by transmitting much more often.

MAP-XAI-only isolates our packetization contribution. Relative to it, full PRISM has statistically tied critical-event recall, coverage, and fairness in all 12 dataset/XAI/privacy settings; traffic is lower in five and tied in seven, while Top-K is tied in 11 and higher in one. Across all 120 PRISM runs there are no privacy-budget violations; the minimum critical-event coverage is 5.98% and minimum Jain fairness is 0.972, above the configured 5% and 0.88 guardrails.

Table II exposes the representative $\epsilon_{\max} = 8$ operating point without dataset-level macro-averaging. The communication savings persist on both OPSD and UCI, while the much higher AoX-DP coverage remains explicit. This prevents the aggregate result from masking the central service trade-off: the proposed stack is designed for selective, high-value explanation delivery rather than exhaustive critical-event service.

Figure 3 supplies the temporal evidence behind the aggregate results: selective private modes spend their finite release allowance early and then accumulate AoX, whereas AoX fixed-DP continues transmitting. Table III separates statistical robustness from raw means. Against the external baselines, traffic is lower in every paired setting, and Top-K is usually indistinguishable or better. Against the internal MAP-XAI-only restriction, the intended result is compatibility rather than dominance: coverage, recall, and fairness are tied in all 12 settings while the full controller retains access to additional modes.

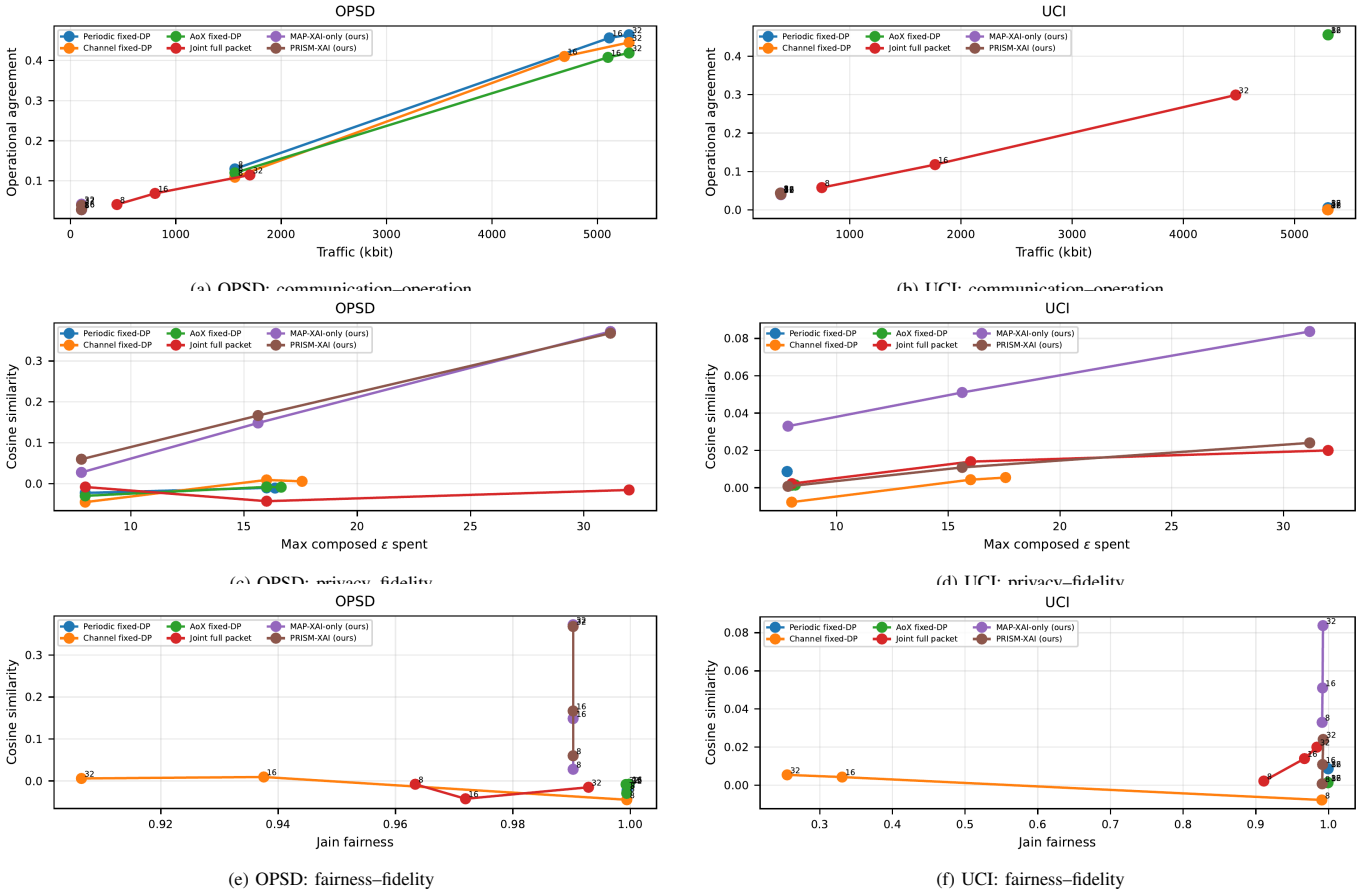
B. Where explanation fidelity is lost

Figure 4 gives the central diagnostic result. Clipping is almost lossless: Top-K changes from 1.000 to 0.990 on OPSD and 0.993 on UCI. The large collapse occurs when cumulative Gaussian DP is added: Top-K becomes 0.241 and 0.282, respectively, close to the random Top-K reference of 0.25. Applying Top-k/quantization after DP changes these values only to 0.252 and 0.295. Hence semantic compression is not

TABLE I

MACRO-AVERAGE OVER 12 DATASET/XAI/PRIVACY SETTINGS (10 PAIRED SEEDS EACH). MAP-XAI-ONLY IS A PROPOSED-COMPONENT ABLATION, NOT AN EXTERNAL BASELINE. LOWER TRAFFIC AND KBIT/CORRECT ARE BETTER; HIGHER TOP-K, COVERAGE, RECALL, PRECISION, AND FAIRNESS ARE BETTER.

Method	Traffic (kbit)	Top-K	Crit. cov.	Crit. recall	Precision	kbit/correct	Fairness
Periodic fixed-DP	4645.2	0.252	0.383	0.178	0.226	707.0	1.000
Channel fixed-DP	4573.8	0.264	0.334	0.161	0.189	1146.6	0.737
AoX fixed-DP	4643.6	0.255	0.875	0.386	0.285	23.6	0.999
Joint full packet	1655.4	0.260	0.248	0.117	0.314	25.5	0.965
MAP-XAI-only (ours)	243.7	0.265	0.078	0.037	**0.538**	**10.0**	0.991
PRISM-XAI (ours)	**242.5**	**0.279**	0.078	0.037	0.510	**10.0**	0.991



(a) OPSD: communication–operation (b) UCI: communication–operation

(c) OPSD: privacy–fidelity (d) UCI: privacy–fidelity

(e) OPSD: fairness–fidelity (f) UCI: fairness–fidelity

Fig. 2. **Main v19 evidence.** Rows show communication–operation, privacy–fidelity, and fairness–fidelity trade-offs. Curves average both XAI implementations and 10 paired seeds at $\epsilon_{\max} \in \{8, 16, 32\}$. MAP-XAI-only is our restricted packetization layer, not an external baseline. The proposed stack occupies a low-traffic, high-fairness selective-release region; higher external-baseline coverage requires substantially more traffic.

the main cause of the observed fidelity limit. Once privacy noise has removed the signal, a wireless scheduler cannot reconstruct it.

The privacy–release sweep confirms why selective release is useful. At $\epsilon = 32$, UCI Top-K is 0.488 with $M = 4$ private releases but falls toward 0.284 at $M = 64$ while coverage grows from 10.3% to 53.5%. On OPSD, $M = 16$ gives Top-K 0.375 at 31.1% coverage, whereas $M = 64$ gives Top-K 0.287 at 83.0% coverage. Figure 5 shows this feasibility boundary. Higher coverage is possible, but spending a fixed household privacy budget over more releases increases the noise per release and can reduce explanation fidelity. This is the reason for the M8 selective-release operating point in the main experiment.

At $\epsilon = 32$, OPSD moves from 8.4% critical coverage and Top-K 0.377 at $M = 4$ to 83.0% coverage and Top-K 0.287 at $M = 64$; $M = 16$ provides 31.1% coverage with Top-K 0.375. UCI moves from 10.3% coverage/0.488 Top-K at $M = 4$ to 53.5%/0.284 at $M = 64$. Thus coverage can be increased, but not for free under a fixed horizon privacy budget.

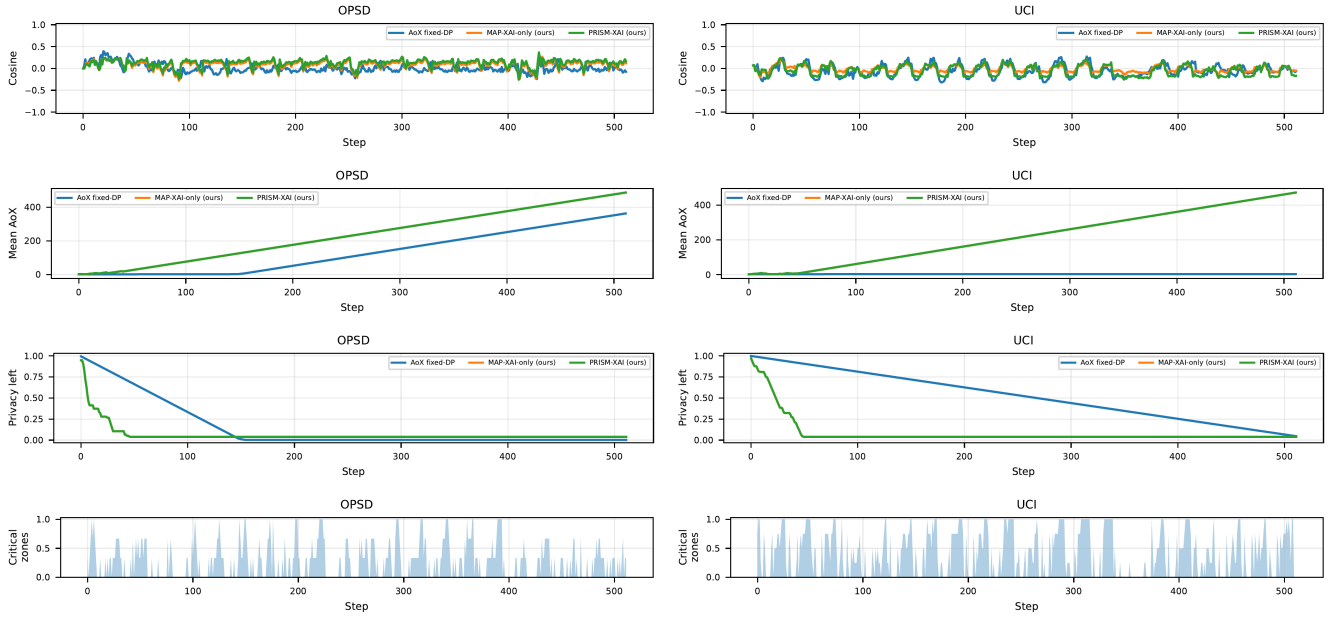
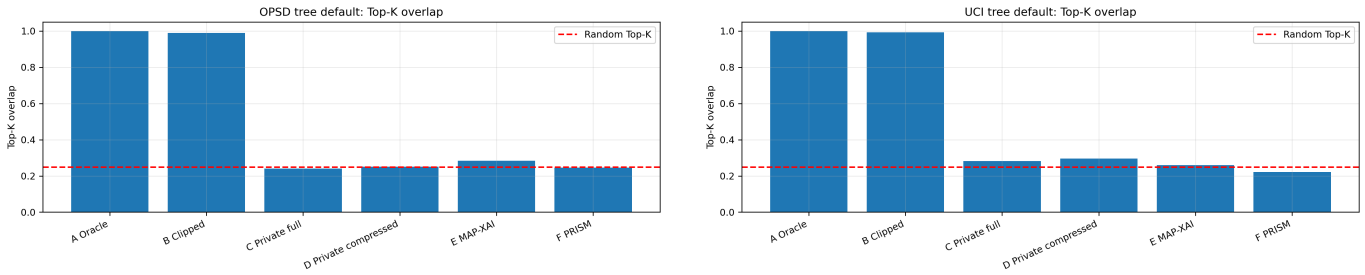


Fig. 3. **Main v19 regime evidence** at $\epsilon_{\max} = 8$, seed 11, averaged over the two XAI implementations. Rows show explanation cosine, mean AoX, remaining privacy fraction, and the public critical-zone process. MAP-XAI-only and PRISM exhaust their deliberately small private-release allowance early and then accumulate AoX; AoX fixed-DP continues releasing and therefore keeps AoX lower at much larger communication cost.



(a) OPSD (b) UCI

Fig. 4. Feasibility ladder at $\epsilon_{\max} = 4$. Clipping preserves Top-K almost perfectly; the dominant collapse occurs when cumulative household-level DP is introduced. The scheduler stages are shown only to complete the mechanism ladder, not as an external PRISM-versus-MAP contest.

TABLE II
REPRESENTATIVE $\epsilon_{\max} = 8$ RESULTS, AVERAGED OVER THE TWO ATTRIBUTION METHODS AND 10 PAIRED SEEDS. MAP IS THE PROPOSED MAP-XAI-ONLY RESTRICTION.

Data	Method	Traffic (kbit)	Crit. cov.	Crit. recall	Precision	kbit/correct
OPSD	AoX fixed-DP	1560.6	0.279	0.120	0.202	34.9
	Joint full	441.4	0.084	0.041	0.267	32.1
	MAP-XAI-only	106.2	0.071	0.028	0.303	9.9
	PRISM-XAI	**104.6**	0.071	0.028	**0.306**	**9.7**
UCI	AoX fixed-DP	5300.2	0.997	0.456	0.362	14.1
	Joint full	748.4	0.130	0.058	0.398	16.2
	MAP-XAI-only	381.4	0.090	0.044	0.488	10.9
	PRISM-XAI	**376.2**	0.090	0.044	**0.493**	**10.7**

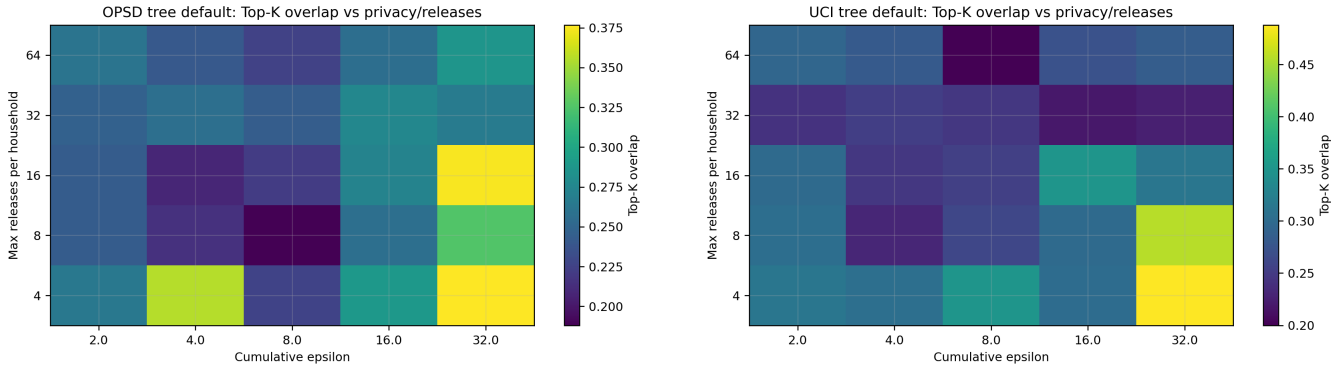
TABLE III
PAIRED 95% CI OUTCOMES ACROSS MATCHED SETTINGS. "BETTER" MEANS THE PRISM-MINUS-REFERENCE CONFIDENCE INTERVAL IS STRICTLY FAVORABLE; "TIE" CONTAINS ZERO. MAP-XAI-ONLY IS AN INTERNAL ABLATION.

Reference	n	Traffic lower	kbit/correct lower	Top-K: B/T/W
Fixed-DP schedulers	36	36	34	6/27/3
Joint full packet	12	12	11	2/9/1
MAP-XAI-only (ours)	12	5	0	1/11/0

C. Interpretation and limitations

The results support a communications contribution: a privacy-aware semantic packet layer plus adaptive scheduling can sharply reduce operator-facing XAI traffic while respecting cumulative household privacy and endpoint fairness. They do *not* support the claim that high-recall automated intervention is solved. The M8 policy is conservative; its mean critical-event recall is 3.7%. Low false intervention and high precision must therefore be interpreted together with the 7.8% mean critical-event coverage. The release sweep shows how coverage can be increased, but also why doing so under a fixed full-horizon privacy budget can destroy attribution signal.

Further limitations are the synthetic zone hierarchy and



(a) OPSD (b) UCI

Fig. 5. Privacy-release sweep: Top-K versus the allowed private-release cap for $\epsilon \in \{2, 4, 8, 16, 32\}$. More releases increase service opportunities, but the same horizon privacy budget must be spread across more releases, so fidelity is non-monotone. Full cosine, recall, traffic, and privacy curves are reported in the accompanying reproducibility appendix.

the scheduler-level wireless abstraction. The privacy theorem protects packet contents under the stated household adjacency, not arbitrary traffic-analysis side channels. Physical feeder topology, link-level or field traces, and event-window/daily privacy accounting are important next validation steps. The present contribution is the joint design and its measured feasibility boundary, not a claim of deployment-complete grid control.

a) Extended analysis.: A reader-facing supplementary report contains the complete 72-condition v19 M8 means, all paired comparisons, M4–M8 sensitivity, temporal-regime analysis, the full feasibility ladder, the 50-setting privacy–release sweep, validation metadata, and result-by-result discussion. It is not required for any claim made here and can be provided by the authors upon request.

VI. Conclusion

We presented PRISM-XAI as a two-layer private explanation-delivery framework rather than a competition between two proposed methods. MAP-XAI provides the privacy-accounted semantic representation; PRISM-XAI schedules those modes under wireless, deadline, freshness, criticality, fairness, and remaining-privacy constraints. In 720 paired runs, the proposed stack has no privacy-budget violations and reduces traffic by 79.4–94.6% relative to external full/fixed-DP policies at a conservative selective-release operating point. The feasibility ladder identifies cumulative household-level DP as the dominant explanation bottleneck, while compression contributes little additional loss. This finding changes the design priority: future gains should come from careful private-release timing and privacy-horizon design, not simply from more aggressive explanation compression or scheduler retuning.

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